

Cycles in the 3-Big Pancake graphs

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Cayley graphs

Let G be a discrete group and S be a set with the property: $S = S^{-1}$, that is $\forall s \in S \exists s^{-1} \in S$ and $e \notin S$.

Cayley graph $\text{Cay}(G, S)$ of the group G with the generating set S is a graph whose vertices are elements of the group G and the element g is connected by an edge exactly to those elements that are obtained by multiplying g by elements from S .

The Hamiltonianity of the Cayley graphs

Conjecture 1

Every connected Cayley graph contains a Hamiltonian cycle.

Conjecture 2 (L. Babai)

For any $\varepsilon > 0$ there exist infinitely many connected vertex-transitive graphs $\Gamma = (V, E)$ (in particular the Cayley graphs), which do not contain cycles of length $l \geq (1 - \varepsilon)|V|$.

Prefix reversals

Let n be a positive integer, $n \geq 3$.

Let Sym_n be the symmetric group of order n .

Permutation $r_i = [i(i-1) \dots 1(i+1) \dots n]$, $2 \leq i \leq n$, which changes the order of the elements inside the segment $[1, i]$ of permutation π on the inverse with left multiplication

$$r_i \cdot [\pi_1 \pi_2 \dots \pi_{i-1} \pi_i \pi_{i+1} \dots \pi_n] = [\pi_i \pi_{i-1} \dots \pi_2 \pi_1 \pi_{i+1} \dots \pi_n]$$

is called a **prefix reversal**.

Denote by PR the set of all prefix reversals of Sym_n .

Pancake graph

The graph $P_n = \text{Cay}(\text{Sym}_n, PR)$ is called the **Pancake graph** of order n .

The forms of the cycle in Pancake graph

A sequence of prefix reversals $C_k = r_{i_0} \dots r_{i_{k-1}}$, $i_j \neq i_{j+1}$, such that $r_{i_0} \dots r_{i_{k-1}} = id$, is called **a form of a cycle** C_k of length k .

The canonical form C_k of an k -cycle is called a form with a lexicographically maximal sequence of indices $i_0 \dots i_{k-1}$.

Cycles in Pancake graph

Theorem 1 (Sheu J.J., Tan J.J.M., Chu K.T., 2006)

There are cycles of length k in Pancake graph P_n , where $6 \leq k \leq n!$.
There are no cycles of length 3, 4 и 5 in Pancake graph.

Greedy sequences

Let $GR = \{r_{m_1}, r_{m_2}, \dots, r_{m_k}\} \subseteq PR$, $k \leq n - 1$ be the ordered set of prefix reversals.

Greedy Hamiltonian cycle is the Hamiltonian cycle formed by the sequential usage of the prefix reversal $r_{m_i} \in GR$ with the smallest possible index i at each step. The sequence GR is called **the greedy sequence**.

Greedy Hamiltonian cycles

There are two known Greedy Hamiltonian cycle constructions: Zachs's construction $GR_Z = \{r_2, r_3, \dots, r_{n-1}, r_n\}$ (1984) and Williams's construction $GR_W = \{r_n, r_{n-1}, \dots, r_3, r_2\}$ (2013).

3-Big Pancake graph

The 3-Big Pancake graph is a Cayley graph on the symmetric group Sym_n , $n \geq 4$ with the generating set $big_3 = \{r_n, r_{n-1}, r_{n-2}\}$ of three biggest prefix reversals.

Conjecture 3 (J. Sawada, A. Williams, 2016)

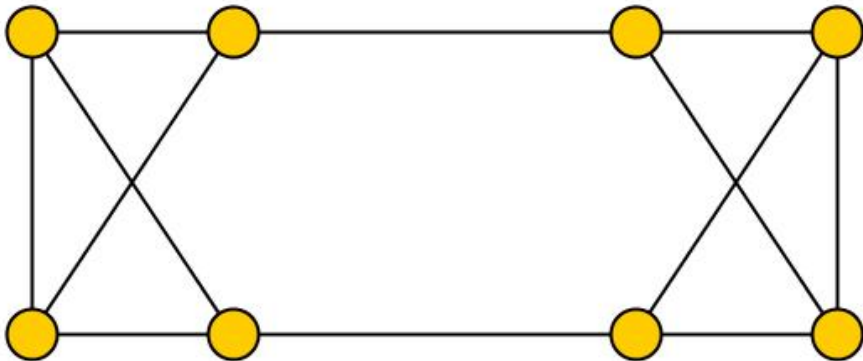
Every 3-Big Pancake graph contains a Hamiltonian cycle.

The spectrum of the cycle in 3-Big Pancake graph

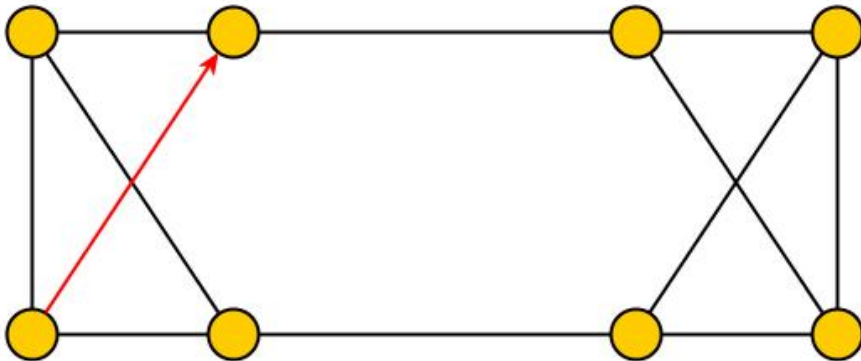
For any Hamiltonian cycle, written as a sequence of prefix reversals, we define **the spectrum of this Hamiltonian cycle** as the sequence of numbers of each prefix reversal's entries.

For example, for the Hamiltonian cycle in Pancake graph P_4 (is also the 3-Big Pancake graph) generated by the Zachs's sequence $(r_2 r_3 r_2 r_3 r_2 r_4)^4$, the spectrum is $r_4 - 4, r_3 - 8, r_2 - 12$.

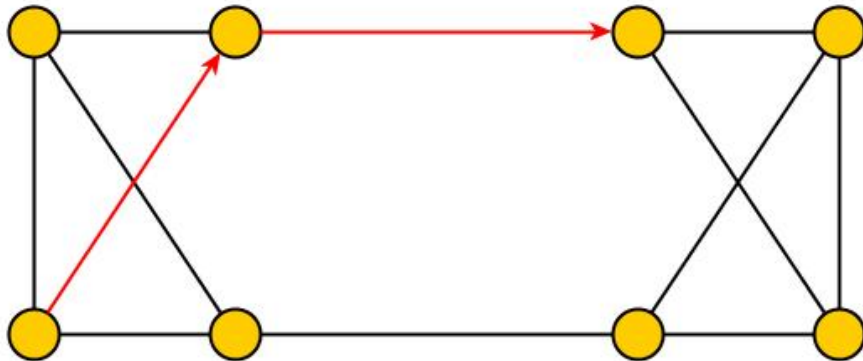
Cutting tail algorithm



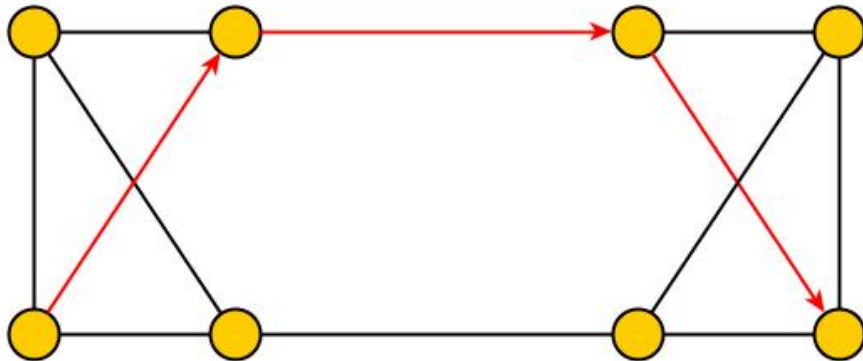
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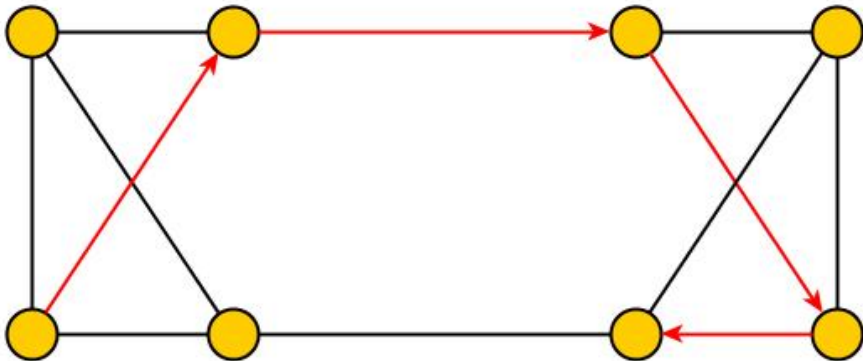
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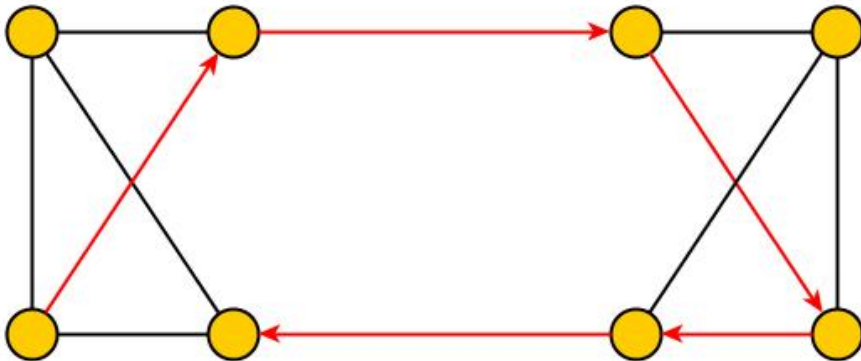
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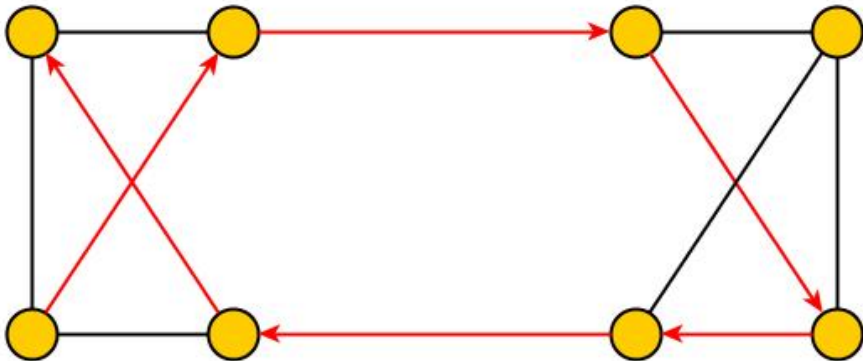
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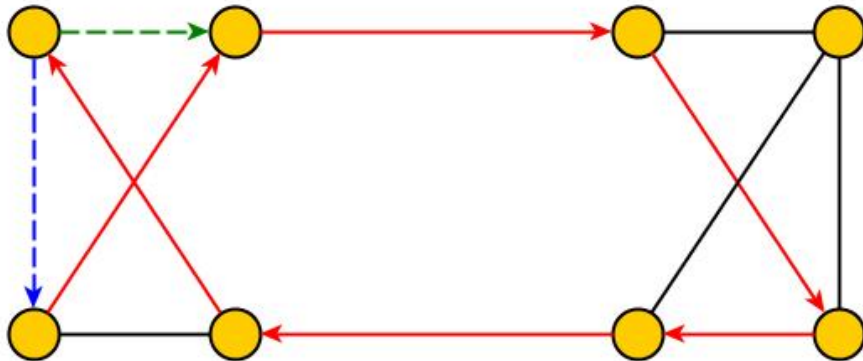
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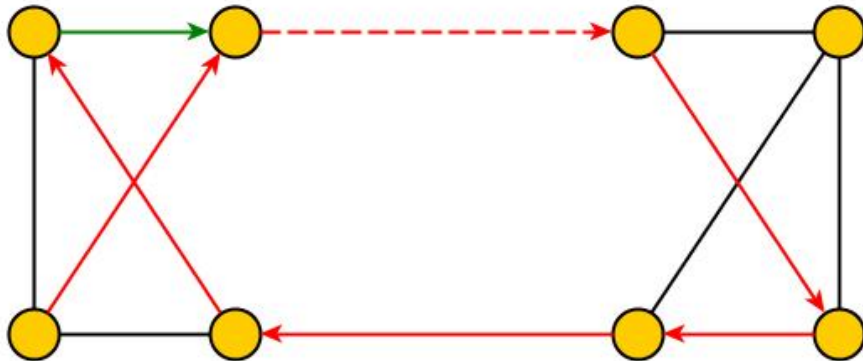
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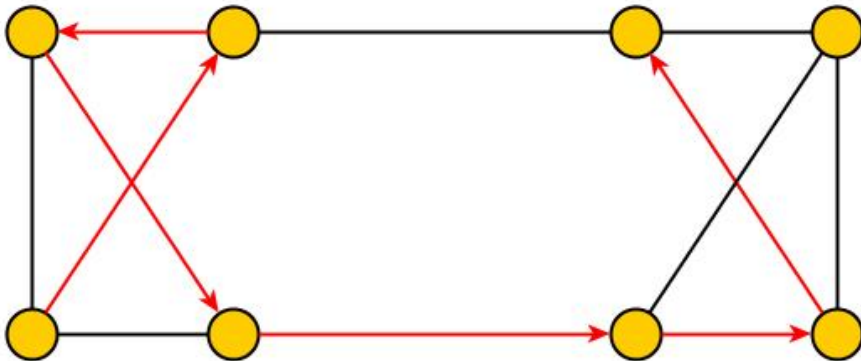
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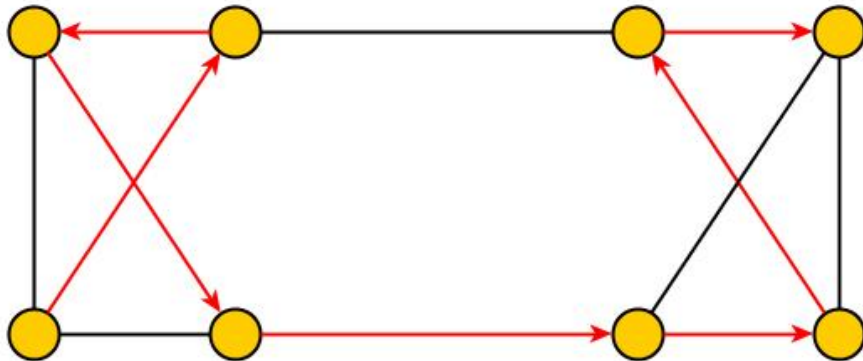
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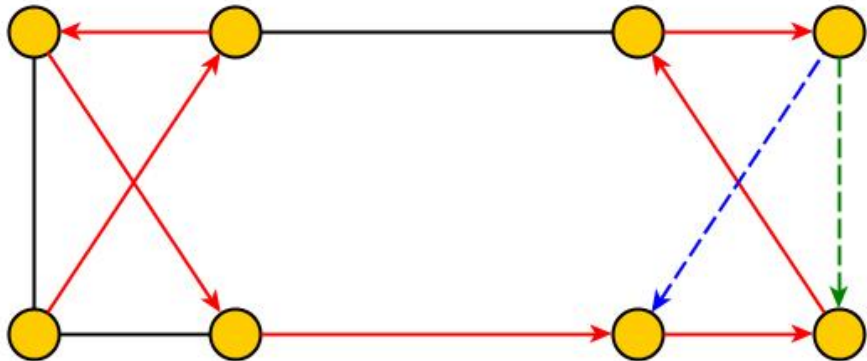
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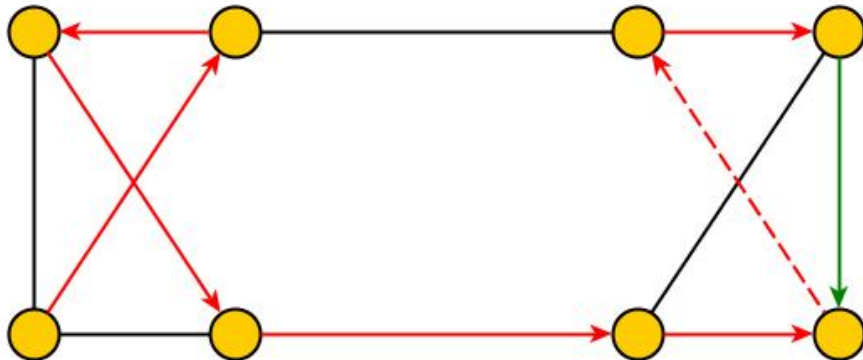
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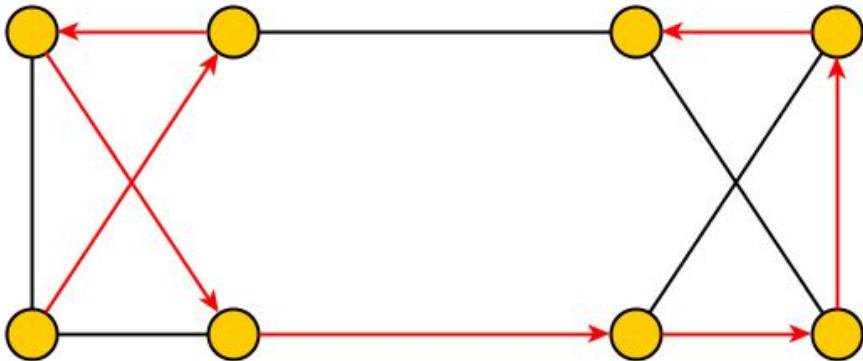
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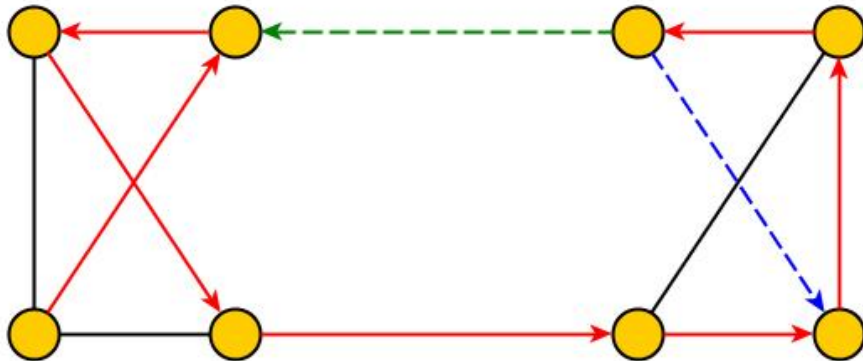
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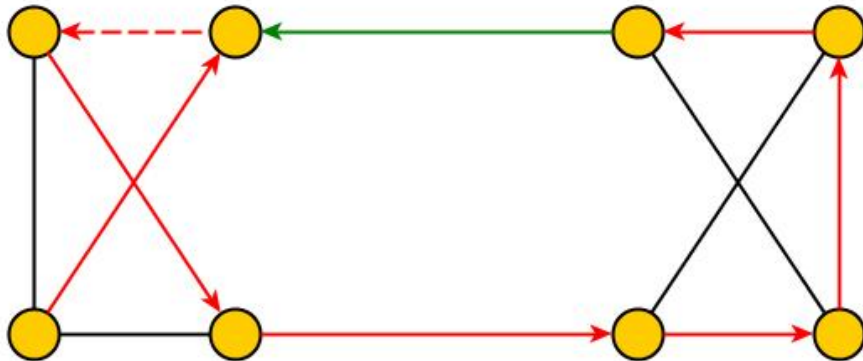
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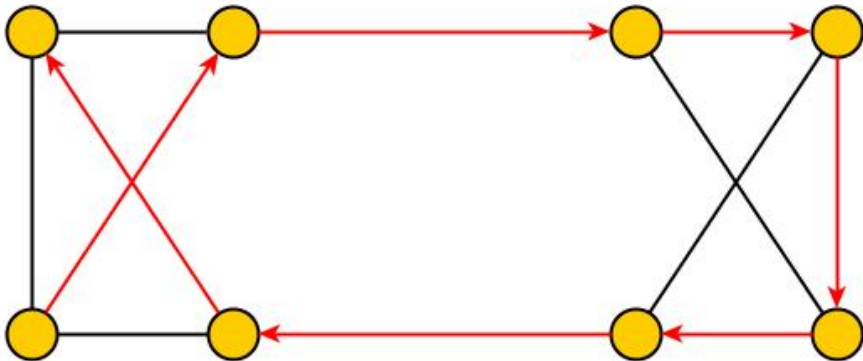
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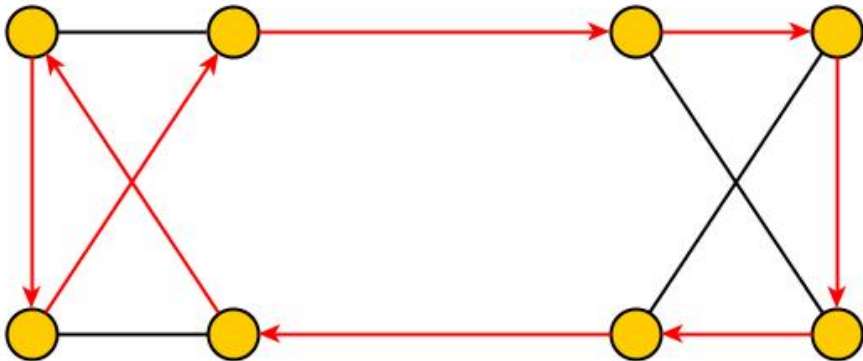
Cutting tail algorithm



Cutting tail algorithm



Cutting tail algorithm



Hamiltonian cycles in 3-Big Pancake graph

After multiple launches of the algorithm, we found many different spectra of Hamiltonian cycles in P_n^{big3} for $n = 4, 5, 6, 7, 8$. So conjecture 3 is correct for $n = 4, 5, 6, 7, 8$.

Every found cycle was not periodic and did not contain visible regularities.

Spectra of Hamiltonian cycles in 3-Big Pancake graph of order 5

(24,48,48)	(31,41,48)	(36,38,46)	(41,37,42)
(26,46,48)	(31,43,46)	(36,40,44)	(41,39,40)
(27,45,48)	(31,49,40)	(36,42,42)	(41,45,34)
(27,49,44)	(32,38,50)	(38,40,42)	(41,49,30)
(28,42,50)	(32,40,48)	(38,42,40)	(42,38,40)
(28,44,48)	(32,44,44)	(39,39,42)	(43,31,46)
(28,46,46)	(33,45,42)	(39,41,40)	(43,35,42)
(28,48,44)	(34,40,46)	(39,43,38)	(43,39,38)
(29,43,48)	(35,33,52)	(40,36,44)	(45,29,46)
(29,45,46)	(35,39,46)	(40,38,42)	(47,31,42)
(30,46,44)	(35,41,44)	(40,40,40)	(50,32,38)

Cycles in 3-Big Pancake graph

Theorem 2

There are cycles of length 6, 7, 8 or 9 given by eleven distinct canonical forms in $P_n^{big_3}$ for $n = 4, 5$.

There are no cycles of length 6, 7 or 9 in $P_n^{big_3}$ for $n \geq 6$.

Theorem 3

Each of vertices of $P_n^{big_3}$, $n \geq 6$, belongs to four distinct 8-cycles of the following canonical form:

$$C_8 = r_n r_{n-1} r_{n-2} r_{n-1} r_n r_{n-1} r_{n-2} r_{n-1}.$$

There are $\frac{n!}{2}$ distinct 8-cycles in $P_n^{big_3}$, $n \geq 6$.

Cycles in 3-Big Pancake graph

Theorem 4

The 3-Big Pancake graph $P_n^{big_3}$ contains cycles of the following length:

- 1) $2n$, $2(n-1)$, and $2n^2 - 2n$ for any $n \geq 4$;
- 2) $2n^2$ for any even $n \geq 4$;
- 3) $\frac{n^2-1}{2}$, $\frac{n^3-n}{2}$ and $\frac{n^3-n^2-n+1}{2}$, for any odd $n \geq 5$
- 4) $\frac{8n}{3}$ for any $n \geq 6$, where 3 divides n .

Cycles in 3-Big Pancake graph

Greedy sequence $\{r_{n-2}, r_{n-1}\}$ generates cycle $C_{2(n-1)}$.

Greedy sequence $\{r_{n-1}, r_n\}$ generates cycle C_{2n} .

Greedy sequence $\{r_{n-2}, r_n\}$ generates cycle C_{2n} (when n is even) or $C_{\frac{n^2-1}{2}}$ (when n is odd).

Cycles in 3-Big Pancake graph

Greedy sequence $\{r_n, r_{n-1}, r_{n-2}\}$ generates cycle C_{2n^2-2n} .

Greedy sequence $\{r_{n-2}, r_n, r_{n-1}\}$ generates cycle C_{2n^2} (when n is even) or $C_{\frac{n^3-n}{2}}$ (when n is odd).

Greedy sequence $\{r_{n-1}, r_{n-2}, r_n\}$ generates cycle C_{2n^2-2n} (when n is even) or $C_{\frac{n^3-n^2-n+1}{2}}$ (when n is odd).

Cycle $C_{\frac{8n}{3}}$ can be written as $(r_{n-1} r_n r_{n-2} r_n)^{\frac{2n}{3}}$.

Thanks for your attention!